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1.0 Introduction

The healthcare industry is rapidly evolving with medical imaging becoming one of the most important things for early detection and diagnosis of diseases. Magnetic Resonance Imaging (MRI) is one of the advanced tools used to diagnose patients to let the clinicians get the detailed images of internal body structure like organs, soft tissues and bones to identify tumors, heart problems and many more. However, as the volume of patients increases, more magnetic resonance imaging is needed and this gives pressure to medical professionals to accurately derive insights from complex information in a timely manner.

This project's purpose is to develop an AI assisted MRI diagnostic system which is an intelligent solution that is designed to support medical teams to diagnose diseases. By integrating AI with traditional MRI, the new system aims to predict the patient's diseases by combining the visual from the MRI and the medical record from the patient such as symptoms, lab results and also previous diseases to give the best solution for the patients.

This project utilizes Design Thinking methodology to address real-world challenges in the healthcare sector. The proposed solution leverages Knowledge Representation (KR) and State Space Search to simulate medical reasoning, ultimately functioning as a "second pair of eyes" for clinicians to ensure faster, more confident, and accurate decision-making

2.0 Problem Statement

Despite the advanced imaging capabilities of modern MRI technology, the diagnostic process remains heavily reliant on manual interpretation by radiologists and doctors. This dependency creates significant challenges because the rapidly increasing volume of medical data often overwhelms healthcare professionals. This situation leads to bottlenecks in patient care and potential inaccuracies in diagnosing complex conditions.

The specific problems identified in the current workflow are:

- **High Risk of Human Error due to Fatigue:** Radiologists analyze thousands of image slices daily which leads to cognitive overload and visual fatigue. This significantly increases the likelihood of missing subtle abnormalities or misinterpreting complex spinal conditions.
- **Inefficiency in Time-Sensitive Diagnoses:** The manual review process is labor-intensive and time-consuming. This creates delays in the diagnostic workflow and prevents patients with urgent conditions such as aggressive tumors from receiving immediate treatment.
- **Difficulty in Differentiating Similar Pathologies:** Certain spinal conditions such as slipped discs and early-stage tumors can present with visually similar patterns on MRI scans. Distinguishing between them often requires simultaneous cross-referencing with patient history which is difficult to do manually under time pressure.
- **Disconnection of Patient Data:** Critical clinical data including symptoms, lab results, and history is often stored separately from imaging data. This forces doctors to mentally integrate these sources and leads to fragmented decision-making.

3.0 AI Solution

The proposed AI-Assisted MRI Diagnostic System integrates AI with MRI machines to support doctors in diagnosing chronic diseases. The system performs the following functions:

1. Medical Image Analysis:
AI models automatically analyze MRI scans to detect abnormalities such as tissue inflammation, lesions, tumors, or structural deformities.
2. Symptom & Data Integration:
The system processes patient records (symptoms, lab results, medical history) and compares them with MRI findings using prediction models.
3. AI Diagnostic Engine:
The engine classifies possible chronic diseases with overlapping symptoms (e.g., autoimmune disorders, neurological diseases, diabetic complications), providing probability-based suggestions.
4. Clinical Decision Support:
The system recommends next steps such as further tests, specialist referral, or imaging angles. It also highlights critical areas directly on MRI images.

Doctor–AI Interaction (Chatbot):

Doctors can ask the AI questions about findings, view summaries, or request deeper explanations through a chatbot interface.

4.0 Goal of AI Solution

An AI assistant MRI diagnostic system that uses machine learning and prediction models to identify possible chronic diseases based on medical image data and patient records.

- The system can highlight abnormal areas in MRI images.
- Provide probability based on predictions of possible diseases.
- Offer recommendations for further tests or treatments.
- Could include a chatbot interface for doctors to ask questions about findings.

5.0 Process of Empathize in Design Thinking

User

Doctors, radiologists, and specialists who diagnose chronic diseases using MRI and patient records.

Engagement

Required

- Doctors need to interpret MRI scans correctly and differentiate diseases with similar symptoms.
- They need to make accurate decisions under time pressure.

Observation

- MRI images that may look similar even for different diseases.
- Complex patient data that must be analyzed together.
- Increasing numbers of patients requiring chronic disease evaluation.

Effect

- Spend long hours analyzing MRI scans.
- Conduct multiple tests to confirm diagnosis.
- Discuss difficult cases in medical team meetings.
- Urgent cases needing quicker diagnosis.
- Patients worried about delays.
- Hospitals emphasizing accuracy and efficiency.

Impact

- Doctors feel stressed by the risk of misdiagnosis.
- They worry about missing early symptoms of serious diseases.
- They want a system that helps them make faster and more confident decisions.

6.0 Process of Define in Design

User	Need	Insight
Doctor	To diagnose chronic diseases accurately and faster	Doctors need an intelligent system that can automatically analyze MRI images and patient data to reduce diagnostic time and prevent errors from fatigue or human bias.
Radiologist	To detect abnormalities in medical images efficiently	Radiologists need AI support that highlights suspicious areas in MRI scans so they can focus on verification rather than scanning thousands of image slices manually.
Patient	To receive early and correct diagnosis	Patients need quicker and more reliable diagnostic results so they can begin treatment earlier and improve health outcomes.
Medical Team	To improve workflow efficiency and reduce delays in diagnosis	Hospitals need a system that integrates imaging, lab results, and doctor notes to streamline communication and reduce bottlenecks in diagnosing chronic diseases.

7.0 Knowledge Representation

The proposed AI-assisted MRI Diagnostic System integrates machine learning with medical imaging and patient information to support doctors in identifying potential chronic diseases. To enable intelligent reasoning, decision making, and interpretation of MRI images, the system relies on several forms of logical knowledge representation (KR). This section presents five logical KRs, their formal First-Order Logic (FOL) representation, and how they support achieving the system's overall goal.

KR1 - Abnormality Detection & Data Transmission

Natural Language Statement:

If a sensor detects an abnormality, the sensor sends data about that abnormality.

FOL Representation:

$$\forall s \forall a [\text{Sensor}(s) \wedge \text{Abnormality}(a) \wedge \text{Detects}(s,a) \rightarrow \text{Sends}(s,a)]$$

Predicates used:

The predicate $\text{Sensor}(s)$ indicates that s represents a device capable of monitoring its environment for irregular conditions. The predicate $\text{Abnormality}(a)$ means that a refers to an unexpected or abnormal event detected in the system, such as unusual readings or fault conditions. The predicate $\text{Detects}(s,a)$ shows that sensor s has identified or recognized the abnormality in its surroundings. Finally, the predicate $\text{Sends}(s,a)$ means that sensor s transmits data about the abnormality to the monitoring system so that appropriate action can be taken.

Explanation:

When a sensor observes an abnormality, the rule guarantees that the sensor will immediately send a corresponding data message to the system. This simple cause-and-effect structure keeps the knowledge representation clear, easy to understand, and free from unnecessary complexity. By directly linking the detection event to the sending action, the rule captures the essential behavior needed for automated monitoring without introducing extra predicates or conditions.

How it helps the system goal:

This rule supports the system's goal by ensuring that every detected abnormality is automatically communicated to the monitoring unit, allowing the system to take quick and informed action. Because the data is sent as soon as an issue is detected, the system can maintain continuous awareness of sensor conditions, respond to potential problems earlier, and improve overall reliability and safety.

KR2 - Disease Classification based on Observed Features

Natural Language Statement:

If a patient shows a feature that belongs to a disease, then that disease becomes a possible diagnosis for that patient.

FOL Representation :

$$\forall p \forall d [\text{Disease}(d) \wedge (\exists f (\text{Observed}(p, f) \wedge \text{HasFeature}(d, f))) \rightarrow \text{PossibleDiagnosis}(p, d)]$$

Predicate used:

The predicate $\text{Disease}(d)$ indicates that d represents a known chronic disease stored in the medical knowledge base. The predicate $\text{Observed}(p, f)$ means that patient p exhibits the feature f , which may be a symptom, MRI signal pattern, or visible abnormality. The predicate $\text{HasFeature}(d, f)$ shows that disease d is medically recognized to involve or cause the feature f . Finally, $\text{PossibleDiagnosis}(p, d)$ means that the system considers disease d as a possible condition affecting patient p based on matched features.

Explanation:

This knowledge rule is one of the core logic-based reasoning processes that the AI program goes through to establish a link between individual patients' features and potential diagnoses. This enables the system to match the patient's MRI findings and symptoms with a database of disease characteristics. If a match is found, the AI system ticks that disease as one of the possible illnesses. This is exactly how real doctors operate: they look for similarities between what they observe in patients and what is described in the books. The system uses FOL to identify which diseases match the features observed in the patient's MRI scan or medical records.

How it helps the system goal

First, it can automatically classify diseases by identifying the patient's features and matching them to the disease characteristics. This makes certain that other illnesses with the same MRI patterns are not misdiagnosed. Additionally, it ensures that all diseases fitting the described features are considered in diagnosis. Third, the probability ranking conducted by machine learning to establish the likelihood of each possible disease depends on it. Additionally, it saves physicians a considerable amount of time since they do not need to compare characteristic features manually. Instead, the system will immediately suggest the condition that fits the patient's MRI and symptom profile.

KR3 - Medical Chatbox Interaction and Intelligent Response System

Natural Language Statement:

If the doctor uses the chatbox to ask any question about a disease, MRI finding, or treatment, the system must understand the question, retrieve relevant medical knowledge, analyse any available patient data, and respond with a clear explanation or opinion based on its reasoning rules.

FOL Representation:

$$\forall q \forall d (\text{ChatboxInput}(q) \wedge \text{RelatedTo}(q, d) \rightarrow \\ (\text{UnderstandQuery}(q) \wedge \text{RetrieveInfo}(d) \wedge \text{ProvideGeneralResponse}(q, d)))$$

Predicate used:

- $\text{ChatboxInput}(q)$: Doctor sends a question or message through the chatbox.
- $\text{RelatedTo}(q, d)$: The question is about disease d , treatment d , or MRI finding related to d .
- $\text{UnderstandQuery}(q)$: NLP module interprets the doctor's question.
- $\text{RetrieveInfo}(d)$: System pulls medical facts, guideline data, or disease knowledge.
- $\text{ProvideGeneralResponse}(q, d)$: System gives a general helpful answer.

Explanation:

This will make the AI become like a smart medical chatbot. When the doctor asks any question about the diseases, MRI finding, and treatment, the system will try to understand it and will give a clear explanation or opinion based on search knowledge base and patient data. This is like having an assistant that can answer any question related to the medical instantly.

How it helps the system goal:

It helps the system become interactive and practical because doctors can ask anything they are unsure about and get fast, relevant answers. This speeds up decision making, improves diagnostic confidence, and makes patient care more efficient.

KR 4 - Treatment Of The Diagnosed Diseases

Natural Language Statement:

Every diagnosed disease has at least a treatment

FOL Representation:

$\forall d(\text{DiagnosedDisease}(d) \rightarrow \exists t \text{TreatmentFor}(t, d))$

Predicate Used:

- $\text{DiagnosedDisease}(d)$: d is a diagnosed disease
- $\text{TreatmentFor}(t, d)$: treatment t is used for disease d

Explanation

When the disease is diagnosed, the AI can detect several treatments that can be applied to patients. It states that at least one treatment will be shown. If there is no treatment, it means the disease is new or the treatment has yet to be discovered. This logical rule also helps ensure that the system provides reliable medical support by linking every identified disease to available medical knowledge. It reinforces the idea that diagnosis should always lead to some form of guidance, preventing situations where a patient is left without any recommended action. Additionally, it allows the system to continuously update itself as new treatments emerge, ensuring that the information stays accurate and relevant.

How it helps the system goal:

The system can achieve its main goal of supporting accurate and efficient medical decision-making. It prevents the AI from giving incomplete results by always providing actionable guidance after a diagnosis. This helps doctors or users make faster decisions, improves patient care, and ensures the system remains useful even as medical knowledge grows.

KR5 – Treatment Progress Monitoring and Follow-up Decision

Natural Language Statement:

If a patient follows a treatment and the treatment improves their condition, then the system should update the patient's status and confirm successful recovery. If a patient follows a treatment and the treatment does not improve their condition, the system triggers a follow-up for the patient.

FOL Representation:

$$\forall p \forall t (\text{UnderTreatment}(p, t) \wedge \text{TreatmentImproves}(p, t) \rightarrow \text{UpdateStatus}(p, \text{Recovered}))$$
$$\forall p \forall t (\text{UnderTreatment}(p, t) \wedge \neg \text{TreatmentImproves}(p, t) \rightarrow \text{TriggerFollowUp}(p))$$

Predicates Used:

- $\text{UnderTreatment}(p, t)$: patient p is receiving treatment t .
- $\text{TreatmentImproves}(p, t)$: treatment t is showing positive improvement for patient p .
- $\text{UpdateStatus}(p, \text{Recovered})$: system updates patient p 's status to recover.
- $\text{TriggerFollowUp}(p)$: system suggests further examination, alternative treatment, or specialist consultation.

Explanation:

This rule allows the system to continuously monitor whether the prescribed treatment is effective. If recovery is seen, the system formally updates the patient's health status. If no improvement occurs, it ensures appropriate next steps such as recommending further treatment, modifying medication, or scheduling a follow-up with a specialist. This creates a feedback cycle similar to real medical decision-making.

How it helps the system goal:

This rule enhances the intelligence of the system by adding post-treatment evaluation. It prevents the system from stopping at diagnosis and treatment recommendation, instead supporting continuous monitoring, decision-making, and adaptive care. This improves reliability, ensures better patient outcomes, and supports long-term healthcare assistance.

8.0 Definition of the State Space

Based on the previously defined Knowledge Representation (KR) for medical diagnosis, the problem is formulated as a state space search problem. The KR models medical knowledge such as abnormal detection, symptom matching, disease identification, treatment options, and recovery outcomes. The state space search uses this knowledge to guide the system step-by-step from initial symptom detection to final recovery, ensuring structured and logical medical decision-making.

The state space graph below consists of all possible medical conditions, diagnosis decisions, and treatment outcomes that the AI system may encounter during the diagnostic process. Each node represents a state, and each transition represents an action and decision taken by the system.

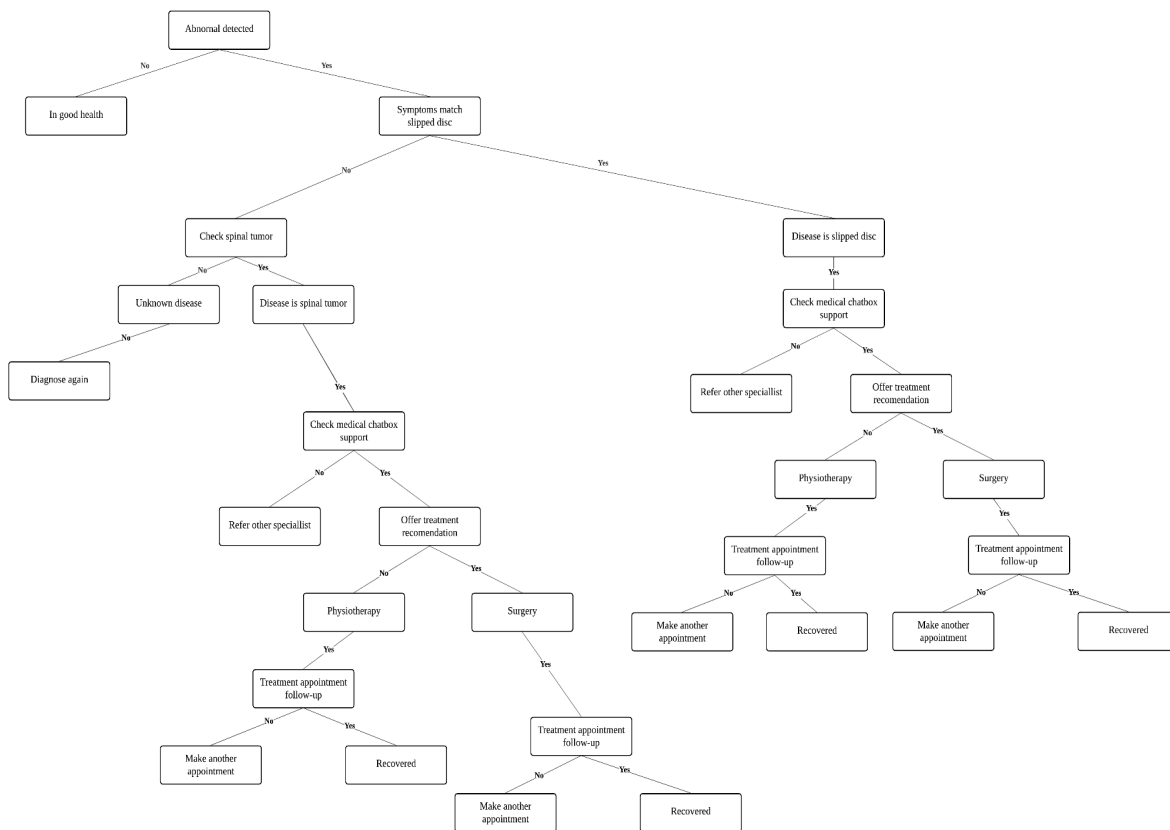


Figure 2.0 MRI System State Space Graph

9.0 Explanation of state and action

9.1 State

No	State	Explanation
1	abnormal_detected	The system checks whether abnormal symptoms are detected from MRI and patient description.
2	in_good_health	No abnormal condition is detected , the patient is healthy.
3	symptoms_match_slipped_disc	The system evaluates whether symptoms match slipped disc characteristics.
4	check_spinal_tumor	Further investigation when symptoms do not match slipped disc.
5	unknown_disease	Disease cannot be identified with current information.
6	diagnose_again	The system requests further diagnosis due to uncertainty.
7	disease_is_slipped_disc	Slipped disc is confirmed as the diagnosis.
8	disease_is_spinal_tumor	Spinal tumor is confirmed as the diagnosis.
9	check_medical_chatbox_support	Checks availability of automated medical support.
10	refer_other_specialist	The patient is referred to a specialist if support is unavailable.
11	offer_treatment_recommendation	The system suggests appropriate treatment options.
12	physiotherapy	Non-surgical treatment is selected.

13	surgery	Surgical treatment is selected.
14	treatment_appointment_follow_up	Patient progress is monitored after treatment.
15	make_another_appointment	Additional treatment sessions are required.
16	recovered	The patient has successfully recovered (goal state).

Table 2.1 shows states and explanation

9.2 Action

1	detect_abnormality	Determines whether abnormal symptoms exist.
2	analyse_symptoms	Matches symptoms against known disease patterns.
3	diagnose_disease	Identifies slipped disc, spinal tumor, or unknown disease.
4	request_re_diagnosis	Initiates further checks when diagnosis is uncertain.
5	provide_chatbox_support	Uses automated medical assistance.
6	refer_specialist	Transfers case to human medical expert.
7	recommend_treatment	Suggests physiotherapy or surgery.
8	schedule_follow_up	Monitors treatment outcome.
9	continue_treatment	Requires another appointment if not recovered.
10	end_process	Ends diagnosis when recovery is achieved.

Table 2.2 shows action and explanation

10.0 Problem Formulation and Explanation

The AI-assisted MRI Diagnostic System is formulated as a state space search problem in which the system systematically evaluates medical conditions to reach an optimal diagnosis and treatment outcome. The state space graph models the diagnostic workflow using medical knowledge such as abnormal detection, symptom matching, disease confirmation, treatment selection, and recovery monitoring. By representing the problem in this manner, the system can move logically from one state to another, ensuring accurate decision-making while reducing uncertainty in medical diagnosis.

10.1 Initial State

The initial state of the system is `abnormal_detected`, which represents the starting point of the diagnostic process. At this state, the system analyses MRI images and patient descriptions to determine whether abnormal symptoms are present. If no abnormality is detected, the system transitions directly to the `in_good_health` state and terminates the process. If an abnormality is detected, the system proceeds to further symptom analysis, ensuring that only patients who require medical attention enter the diagnostic decision process.

10.2 Actions

Actions in the state space represent the medical decisions and system operations that cause transitions between states. Each action reflects a real-world medical decision and ensures that the system progresses logically toward treatment and recovery.

- `detect_abnormality`
- `analyse_symptoms`
- `diagnose_disease`
- `request_re_diagnosis`
- `provide_chatbox_support`
- `refer_specialist`
- `recommend_treatment`

- schedule_follow_up
- continue_treatment
- End_process

10.3 Goal

The goal of the state space search is to reach the recovered state. This goal is achieved when the system successfully diagnoses the patient's condition, provides appropriate treatment recommendations, and confirms recovery through follow-up assessments. The goal state represents the successful completion of the medical decision-making process, where patient health is restored and no further treatment is required.

10.4 Path Cost

Path cost refers to the total number of state transitions required for the system to reach the goal state from the initial state. Each state transition has a cost of one unit, representing the processing time or decision step taken by the system. Paths with fewer transitions have lower cost and indicate faster diagnosis and treatment, while paths with higher cost reflect more complex cases that require additional diagnosis or follow-up. The system prioritizes lower-cost paths when possible while maintaining medical accuracy and patient safety.

Path	Solution Route	Path Cost
1	abnormal_detected → in_good_health	1 unit
2	abnormal_detected → symptoms_match_slipped_disc → disease_is_slipped_disc → check_medical_chatbox_support → offer_treatment_recommendation → physiotherapy → treatment_appointment_follow_up → recovered	7 units
3	abnormal_detected → symptoms_match_slipped_disc → disease_is_slipped_disc → check_medical_chatbox_support →	7 units

	offer_treatment_recommendation → surgery → treatment_appointment_follow_up → recovered	
4	abnormal_detected → symptoms_not_match → check_spinal_tumor → disease_is_spinal_tumor → check_medical_chatbox_support → treatment → surgery → treatment_appointment_follow_up → recovered	8 units
5	Abnormal_detected → symptoms_not_match → check_spinal_tumor → unknown_disease → diagnose_again	4 units

Table 2.4 Path Solution Routes and Corresponding Path Costs

Peer review on Team Working (TW)

Category	4	3	2	1
Active participation	Routinely provides useful ideas when participating in the group.	Usually provides useful ideas when participating in the group.	Sometimes provides useful ideas when participating in the group.	Rarely provides useful ideas when participating in the group.

MEMBERS/ VOTE	MUHAMMAD MUKHRIT Z AL IMAN BIN MOHD RAFFI	MUHAMMAD NAIM BIN ABDULLAH	AHMAD ADIB ZIKRI BIN A.MAZLAM	PRAVINR AJ A/L SIVABATH I	AHMAD ZIYAAD BIN MOHD ABBAS
MUHAMMAD	-	4	4	4	4

MUKHRITZ AL IMAN BIN MOHD RAFFI					
MUHAMMA D NAIM BIN ABDULLAH	4	-	4	4	4
AHMAD ADIB ZIKRI BIN A.MAZLAM	4	4	-	4	4
PRAVINRAJ A/L SIVABATHI	4	4	4	-	4
AHMAD ZIYAAD BIN MOHD ABBAS	4	4	4	4	-

11.0 PEAS Model Representation

MRI Scanner

Performance	- High accuracy in detecting abnormalities from MRI scans - Correct identification of diseases
Environment	- Patient - Doctor - Hospital
Actuator	- Scan entire body - Flag abnormal findings
Sensor	- MRI image data - Patients medical records

An MRI scanner is a medical machine that takes detailed pictures of the inside of the body to help doctors find and diagnose illnesses. It aims to give clear, accurate images so clinicians can see problems like tumors or injuries, and it can scan parts of the body or the whole body as needed. The results are sent to doctors so they can make the right treatment decisions.

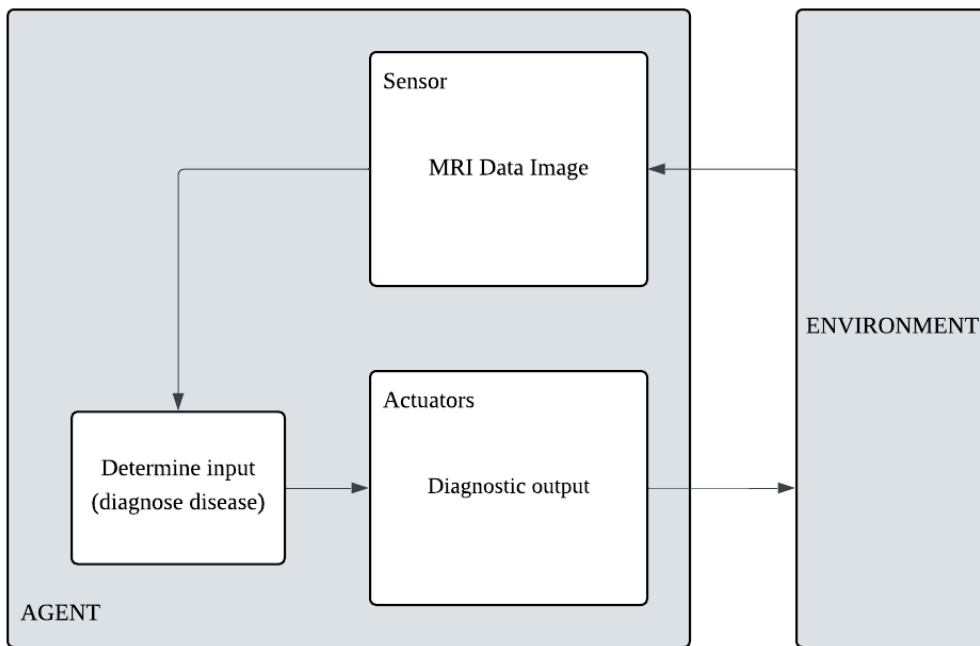
The machine itself collects signals from the body and turns them into images, while motors and controllers move the patient table and run the scan steps. It works in hospitals around patients, radiologists, and technicians, so safety and patient comfort are important. When used properly, an MRI helps doctors diagnose conditions faster and more reliably.

AI Chatbox

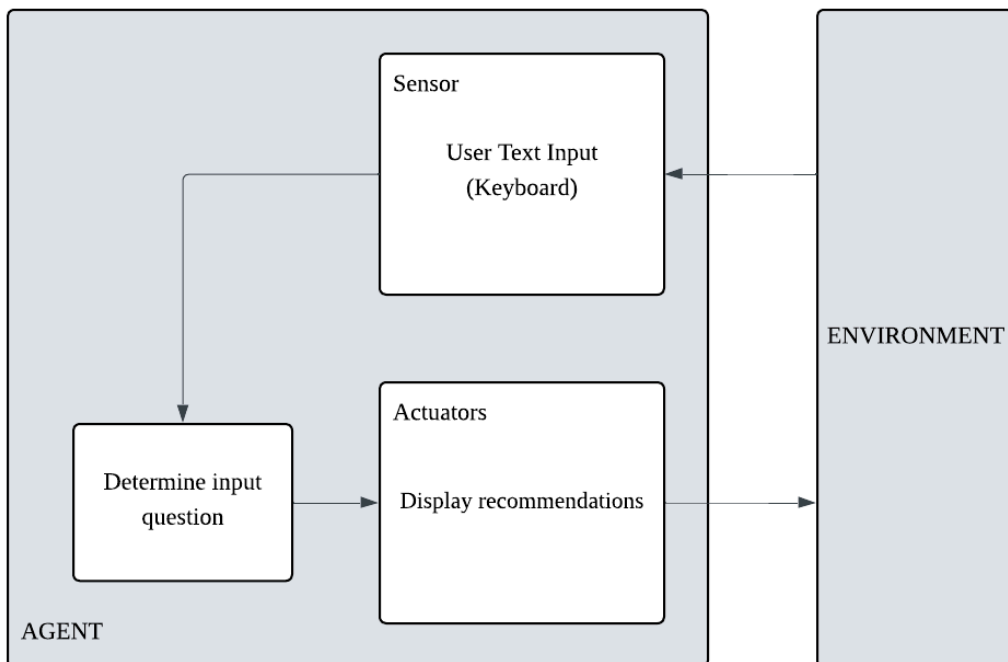
Performance	- Recommendation of treatments - Patient recovery rate
Environment	- Patient - Doctor - Hospital
Actuator	- Interact with ai chatbox - Provide treatment recommendations - Schedule follow-up appointments
Sensor	- User input (keyboard)

An AI chatbot is a tool that has been created to support doctors in diagnosing diseases and recommending suitable treatments. Through the conversations, doctors can consult the AI chatbot to confirm possible diagnoses and clarify uncertain clinical questions. By analyzing medical information efficiently, the system helps reduce the time that is required for diagnosis and supports faster, more informed clinical decision-making, ultimately improving the quality and efficiency of patient care.

Diagram Of Representation MRI Scanner



AI Chatbox



12.0 Explanation in Proof Of Concept (POC)

MRI Scanner

Performance

The MRI Scanner component is intended to provide high-precision diagnostic imaging to assist clinicians in identifying structural abnormalities. The system evaluates its performance based on the clarity of the generated images and the accuracy in detecting specific pathologies such as tumors or tissue injuries. As a result, when a patient undergoes a scan, the system focuses on delivering artifact-free visual data that allows doctors to make the right treatment decisions with confidence. Furthermore, the system aims to enhance the reliability of disease identification by minimizing errors in signal processing. The scanner assures consistent diagnostic quality, ensures patient safety during operation, and contributes to faster diagnosis cycles by producing high-fidelity scans that reduce the need for repeat imaging.

Environment

The environment of the MRI Scanner consists of the patient, doctor, and hospital setting in which the imaging process takes place. The patient is the primary element of the environment, as their physical condition and internal anatomy determine the MRI image data captured by the system. The doctor operates within this environment as the medical professional who relies on the scanner's output to identify abnormalities and make diagnostic decisions. The hospital provides a controlled clinical environment that enforces safety regulations, supplies supporting medical infrastructure, and enables integration with patient medical records, ensuring that the MRI Scanner functions accurately, safely, and reliably within real clinical conditions.

Sensor

The input data collected by the MRI Scanner is critical for generating accurate diagnostic outputs. The primary sensors involved are the Magnetic Signal Receiver and the Patient Record Interface. The Magnetic Signal Receiver captures raw radio frequency signals emitted from the body's tissues and converts them into detailed digital images. Simultaneously, the Patient Record Interface acts as a sensor that retrieves existing medical data, such as previous diagnoses or contraindications, from the hospital database. These sensors are essential to ensure the flow of data into the processing unit, allowing the system to construct a comprehensive view of the patient's internal anatomy for analysis.

Actuator

The MRI scanner's actuators provide both mechanical and system-level functions necessary for image acquisition and output delivery. This encompasses gradient coil controllers, RF transmitters, and patient table motors that regulate magnetic field fluctuations and patient positioning during the scan. Upon completion of processing, the diagnostic output actuator

produces finished MRI pictures and emphasizes identified problems, transmitting them to the clinician's workstation. These actuators transform system decisions into tangible scans and visual outputs, facilitating precise diagnosis and clinical assessment.

AI Chatbox

Performance

The AI Chatbox is intended to function as an intelligent support tool that optimizes the clinical decision-making process for doctors. The system's performance is measured by the relevance of its treatment recommendations and its long-term impact on patient recovery rates. When a doctor consults the system, it analyzes the query to provide evidence-based medical advice, confirming diagnoses and clarifying uncertain clinical scenarios. Furthermore, the system identifies opportunities to improve hospital efficiency by reducing the time required for complex case analysis. The Chatbox assures faster, more informed clinical decisions and ultimately improves the quality of patient care by offering instant, accurate medical responses through a conversational interface.

Environment

The environment of the AI Chatbox includes the patient, doctor, and hospital clinical system where medical decision support is required. The doctor interacts directly with the Chatbox by entering clinical queries and reviewing treatment recommendations, making the doctor a key component of the operational environment. The patient contributes to the environment through their diagnosis, symptoms, and recovery progress, which influence the relevance of the Chatbox's responses. The hospital environment supports the system by providing access to diagnostic results, medical records, and treatment resources, allowing the AI Chatbox to generate recommendations that are consistent with clinical protocols and available healthcare facilities.

Sensor

The input data for the AI Chatbox is primarily gathered through the User Input Interface (Keyboard). This sensor allows doctors to manually enter text-based queries, specific clinical questions, or patient symptoms directly into the system. The text input acts as the trigger for the AI's Natural Language Processing (NLP) module, which interprets the doctor's intent. By capturing these specific inquiries, the sensor ensures that the First-Order Logic (FOL) within the system can retrieve the correct medical knowledge and generate appropriate treatment recommendations or follow-up suggestions.

Actuator

The actuator of the AI Chatbox transmits the system's reactions and recommendations to physicians via the chat interface. Upon evaluating user inquiries through Natural Language Processing and First-Order Logic, the actuator presents diagnostic elucidations, therapeutic

recommendations, and subsequent prompts in a coherent and actionable manner. This enables physicians to promptly utilize the AI's reasoning in clinical decision-making, enhancing efficiency and facilitating precise patient care.

13.0 Behaviour of agent to achieve the system goals

13.1 MRI Scanner

The intelligent agent in the MRI Scanner module operates through a continuous cycle of perception, analysis, and action to achieve the system goal of producing accurate diagnostic images that assist doctors in disease identification while ensuring patient safety.

Initially, the agent continuously observes the environment through its sensors. The Magnetic Signal Receiver captures raw radio frequency signals emitted by the patient's tissues during the MRI scan. At the same time, the Patient Record Interface retrieves relevant medical information such as previous diagnoses, scan history, and safety constraints. These sensor inputs allow the agent to perceive both the patient's physical condition and contextual medical data before and during the scanning process.

After receiving the sensor data, the agent processes the information to determine optimal scanning parameters. It evaluates whether the magnetic signals are clear, checks for noise or artifacts, and verifies that the imaging conditions meet diagnostic standards. Based on predefined medical rules and system constraints, the agent decides if adjustments are required, such as modifying magnetic field gradients, scan duration, or imaging sequences. This reasoning step ensures that the system adapts dynamically to patient-specific conditions rather than using a fixed scanning configuration.

Once a decision is made, the agent activates the actuators to execute the required actions. These include controlling the gradient coils, RF transmitters, and patient table motors to adjust positioning and scanning intensity. After the scan is completed, the agent generates high-quality MRI images and highlights potential abnormalities before delivering the diagnostic output to the clinician's workstation. Through this behavior cycle, the MRI Scanner agent ensures reliable image quality, reduces the need for repeat scans, and supports accurate medical diagnosis.

13.2 AI Chatbox

The intelligent agent in the AI Chatbox module is designed to support clinical decision-making by analyzing doctor queries and generating relevant medical recommendations. Its behavior focuses on improving diagnostic clarity, treatment planning, and follow-up efficiency.

The agent begins by perceiving inputs from the User Input Interface, where doctors enter clinical questions, symptoms, or diagnostic concerns using the keyboard. This textual input acts as the primary sensor data, allowing the agent to understand the doctor's intent and the clinical context of the query. The system may also implicitly reference patient-related data, such as recovery progress or MRI findings, when available.

After collecting the input, the agent applies Natural Language Processing (NLP) techniques to interpret the doctor's query. The interpreted intent is then evaluated using First-Order Logic and medical knowledge rules embedded within the system. Based on this reasoning process, the agent determines the most appropriate response, such as confirming a diagnosis, suggesting treatment options, recommending follow-up appointments, or clarifying uncertainties related to patient recovery.

Once a decision is reached, the agent activates its actuator by displaying structured recommendations through the chat interface. These outputs include clear medical explanations, treatment suggestions, and next-step guidance that the doctor can immediately apply in clinical practice. By completing this perception–reasoning–action cycle, the AI Chatbox enables faster, more informed decisions and enhances the overall quality of patient care.

13.3 Continuous Agent Behaviour to Achieve System Goals

Both the MRI Scanner agent and the AI Chatbox agent operate in a continuous feedback loop of monitoring, decision-making, and action. The MRI Scanner focuses on producing accurate diagnostic data, while the AI Chatbox transforms that data and clinical input into meaningful recommendations. Together, these agents work collaboratively to achieve the system's overall goals which are accurate diagnosis, effective treatment support, and improved patient outcomes through intelligent automation.

14.0 Conclusion

In conclusion, this project successfully demonstrates the design and implementation of an intelligent healthcare system through a Proof of Concept (POC) that integrates an MRI Scanner module and an AI Chatbox module. The system applies fundamental Artificial Intelligence concepts such as intelligent agents, PEAS modeling, sensor-actuator interaction, and rule-based reasoning to address real-world healthcare challenges. By combining diagnostic imaging with intelligent decision support, the proposed solution illustrates how AI can enhance both clinical accuracy and operational efficiency in medical environments.

The MRI Scanner module focuses on generating high-quality diagnostic images by continuously monitoring patient signals and dynamically adjusting scanning parameters. Through intelligent agent behavior, the system ensures reliable image output, reduces scanning errors, and supports clinicians in identifying abnormalities with greater confidence. This module highlights the importance of intelligent perception and action in safety-critical healthcare systems.

Meanwhile, the AI Chatbox module complements the diagnostic process by assisting doctors in interpreting medical information and making informed treatment decisions. By utilizing Natural Language Processing and First-Order Logic, the Chatbox provides relevant medical recommendations, clarifies clinical uncertainties, and supports follow-up planning. This interactive component enhances usability and demonstrates how conversational AI can improve decision-making speed and accuracy in a clinical setting.

Overall, the proposed system achieves its primary objective of validating an AI-based healthcare solution through a functional and interactive POC. The integration of intelligent agents across both modules shows how AI technologies can work collaboratively to improve patient care, support medical professionals, and optimize healthcare workflows. As a future enhancement, the system could be expanded with real-time learning models, larger medical datasets, and deeper integration with hospital information systems to further increase accuracy, adaptability, and clinical impact.

1. LINK VIDEO PRESENTATION

<https://youtu.be/rzdm9H9wyE>

2. LINK PROTOTYPE

https://www.canva.com/design/DAG-Wz0eB90/4IH9UKyGJKFvfveGhAcEBw/edit?utm_content=DAG-Wz0eB90&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton